

APPENDIX A: RIGOROUS DERIVATION OF THE FRIEDMANN EQUATIONS FROM THE PSP METRIC

Mathematical Appendix to the PSP Theory

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Author: Evgeny V. Chernoknizhny

ORCID: 0009-0007-2558-9172

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Abstract

This document presents a rigorous mathematical derivation of the Friedmann equations from the metric of the phase-state parameter cosmological model (PSP). It is shown that the PSP metric, after a change of the time coordinate, reduces to the standard FLRW metric, allowing the full formalism of general relativity to be applied. The Friedmann equations are derived in the phase representation, the connection to the Hubble parameter is established, and it is demonstrated that the expression $H(z) = H_0 \sqrt{1 + \alpha(z/2.5)^2 e^{\beta z}}$ naturally follows from the cycle geometry. The document contains the complete derivation, dimensional analysis, and discussion of the physical interpretation of the results.

Keywords: Friedmann equations, PSP metric, cycle phase, general relativity, cosmology,

Hubble parameter.

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1 Introduction

In the standard cosmological model Λ CDM, the dynamics of the Universe's expansion are described by the Friedmann equations, which follow from the FLRW metric and Einstein's field equations. In the PSP model, time is replaced by the cycle phase $M \in [0, 1]$, and the metric has the form:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1 \quad (1)$$

The purpose of this document is to show that this metric is **equivalent** to the FLRW metric after a change of variables, and consequently, the Friedmann equations naturally follow from the PSP geometry.

2 The PSP Metric and Its Relation to FLRW

2.1 The Standard FLRW Metric

In general relativity, spacetime is described by the Friedmann-Lemaître-Robertson-Walker metric:

$$ds^2 = -c^2 dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right) \quad (2)$$

where:

- t is cosmological time,
- $a(t)$ is the scale factor,
- $k = +1, 0, -1$ is the spatial curvature,
- $d\Omega^2 = d\theta^2 + \sin^2 \theta d\varphi^2$.

2.2 The PSP Metric

The PSP metric:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right) \quad (3)$$

where:

- $M \in [0, 1]$ is the dimensionless cycle phase,
- $\Phi_M = dM/dz$ is the phase flow,
- $R_{\text{shell}}(M)$ is the shell radius.

2.3 Change of Variable

Introduce a new time coordinate t such that:

$$dM = \Phi_M \cdot c dt \quad (4)$$

Then:

$$\frac{1}{\Phi_M^2} dM^2 = \frac{1}{\Phi_M^2} \cdot \Phi_M^2 \cdot c^2 dt^2 = c^2 dt^2 \quad (5)$$

And the metric (3) becomes:

$$ds^2 = -c^2 dt^2 + R_{\text{shell}}^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right) \quad (6)$$

80 **Conclusion:** The PSP metric is **equivalent** to the FLRW metric with the scale factor:

$$a(t) = R_{\text{shell}}(M(t)) \quad (7)$$

This means that the full formalism of GR, including Einstein's field equations, is fully applicable to the PSP model after the change of time coordinate.

3 Components of the Einstein Tensor for the FLRW Metric

85 3.1 The FLRW Metric

For the metric (2), the metric tensor components are:

$$g_{00} = -1, \quad g_{ij} = a^2(t) \gamma_{ij}, \quad \gamma_{ij} = \frac{\delta_{ij}}{1 - kr^2} + r^2 \delta_{ij} \quad (8)$$

3.2 Christoffel Symbols

The non-zero Christoffel symbols are:

$$\Gamma_{ij}^0 = \frac{\dot{a}}{c} a \gamma_{ij}, \quad \Gamma_{0j}^i = \frac{\dot{a}}{c} \delta_j^i, \quad \Gamma_{jk}^i = \Gamma_{jk}^i[\gamma] \quad (9)$$

3.3 Ricci Tensor

90 The Ricci tensor components are:

$$R_{00} = -\frac{3}{c^2} \frac{\ddot{a}}{a} \quad (10)$$

$$R_{ij} = \frac{1}{c^2} (2\dot{a}^2 + a\ddot{a} + 2kc^2) \gamma_{ij} \quad (11)$$

3.4 Scalar Curvature

$$R = \frac{6}{c^2} \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2} \right) \quad (12)$$

3.5 Einstein Tensor Components

The Einstein tensor is defined as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \quad (13)$$

Substituting (10), (11) and (12), we obtain:

95 **Time component:**

$$G_{00} = \frac{3}{a^2} (\dot{a}^2 + kc^2) \quad (14)$$

Space components:

$$G_{ij} = -\frac{1}{a^2} (2a\ddot{a} + \dot{a}^2 + kc^2) \gamma_{ij} \quad (15)$$

4 Einstein's Field Equations and Derivation of the Friedmann Equations

4.1 Stress-Energy Tensor of a Perfect Fluid

100 For a perfect fluid:

$$T_{\mu\nu} = \left(\rho + \frac{p}{c^2} \right) u_\mu u_\nu - \frac{p}{c^2} g_{\mu\nu} \quad (16)$$

In the fluid's rest frame $u^\mu = (c, 0, 0, 0)$:

$$T_{00} = \rho c^2, \quad T_{ij} = \frac{p}{c^2} g_{ij} = pa^2 \gamma_{ij} \quad (17)$$

4.2 Einstein's Field Equations

Einstein's field equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (18)$$

4.3 First Friedmann Equation

105 Substituting (14) and (17) into (18) for $\mu = \nu = 0$:

$$\frac{3}{a^2} (\dot{a}^2 + kc^2) = \frac{8\pi G}{c^4} \cdot \rho c^2 = \frac{8\pi G}{c^2} \rho \quad (19)$$

Multiplying both sides by a^2 :

$$3\dot{a}^2 + 3kc^2 = \frac{8\pi G}{c^2}\rho a^2 \quad (20)$$

Dividing by a^2 :

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3c^2}\rho - \frac{kc^2}{a^2} \quad (21)$$

First Friedmann Equation:

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2}\rho - \frac{kc^2}{a^2}} \quad (22)$$

4.4 Second Friedmann Equation

110 Substituting (15) and (17) into (18) for $\mu = \nu = i$:

$$-\frac{1}{a^2}(2a\ddot{a} + \dot{a}^2 + kc^2) = \frac{8\pi G}{c^4} \cdot pa^2 \cdot \frac{1}{a^2} = \frac{8\pi G}{c^4}p \quad (23)$$

Multiplying by $-a^2$:

$$2a\ddot{a} + \dot{a}^2 + kc^2 = -\frac{8\pi G}{c^4}pa^2 \quad (24)$$

Using the first equation (22), we obtain:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}\left(\rho + \frac{3p}{c^2}\right) \quad (25)$$

Second Friedmann Equation:

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}\left(\rho + \frac{3p}{c^2}\right)} \quad (26)$$

5 Transition to the Phase Representation

115 5.1 Phase Derivatives

Using (4), express time derivatives in terms of phase derivatives:

$$\frac{da}{dt} = \frac{da}{dM} \cdot \frac{dM}{dt} = \frac{da}{dM} \cdot \Phi_{Mc} \quad (27)$$

$$\frac{d^2a}{dt^2} = \Phi_{Mc}^2 \frac{d^2a}{dM^2} + \Phi_{Mc} c^2 \frac{d\Phi_M}{dM} \frac{da}{dM} \quad (28)$$

5.2 First Equation in Phase Representation

Substituting (27) into (22):

$$\left(\frac{1}{a} \cdot \frac{da}{dM} \cdot \Phi_M c\right)^2 = \frac{8\pi G}{3c^2} \rho - \frac{kc^2}{a^2} \quad (29)$$

$$\Phi_M^2 c^2 \left(\frac{1}{a} \frac{da}{dM}\right)^2 = \frac{8\pi G}{3c^2} \rho - \frac{kc^2}{a^2} \quad (30)$$

Canceling c^2 :

$$\boxed{\Phi_M^2 \left(\frac{1}{a} \frac{da}{dM}\right)^2 = \frac{8\pi G}{3c^2} \rho(M) - \frac{k}{a^2}} \quad (31)$$

120 **First Friedmann Equation in phase representation (PSP).**

5.3 Second Equation in Phase Representation

Substituting (27) and (28) into (26):

$$\frac{\Phi_M^2 c^2}{a} \frac{d^2 a}{dM^2} + \frac{\Phi_M c^2}{a} \frac{d\Phi_M}{dM} \frac{da}{dM} = -\frac{4\pi G}{3c^2} \left(\rho + \frac{3p}{c^2}\right) \quad (32)$$

After simplification:

$$\boxed{\Phi_M^2 \frac{1}{a} \frac{d^2 a}{dM^2} + \Phi_M \frac{d\Phi_M}{dM} \frac{1}{a} \frac{da}{dM} = -\frac{4\pi G}{3c^4} (\rho c^2 + 3p)} \quad (33)$$

Second Friedmann Equation in phase representation (PSP).

125 5.4 Continuity Equation

From the conservation of the stress-energy tensor $\nabla_\mu T^{\mu\nu} = 0$:

$$\dot{\rho} + 3\frac{\dot{a}}{a} \left(\rho + \frac{p}{c^2}\right) = 0 \quad (34)$$

In phase representation:

$$\Phi_M c \frac{d\rho}{dM} + 3\Phi_M c \frac{1}{a} \frac{da}{dM} \left(\rho + \frac{p}{c^2}\right) = 0 \quad (35)$$

Canceling $\Phi_M c$:

$$\boxed{\frac{d\rho}{dM} + 3\frac{1}{a} \frac{da}{dM} \left(\rho + \frac{p}{c^2}\right) = 0} \quad (36)$$

6 Connection to the Hubble Parameter in the PSP Model

130 6.1 The Hubble Parameter

In standard cosmology:

$$H(t) = \frac{\dot{a}}{a} \quad (37)$$

In the PSP model, using (27):

$$H(M) = \frac{1}{a} \frac{da}{dt} = \frac{1}{a} \frac{da}{dM} \cdot \Phi_M c \quad (38)$$

$$\boxed{H(M) = \Phi_M(M) \cdot \frac{c}{R_{\text{shell}}(M)} \cdot \frac{dR_{\text{shell}}}{dM}} \quad (39)$$

6.2 Connection to Redshift

In standard cosmology:

$$1 + z = \frac{a_0}{a} \quad (40)$$

135 This relation is preserved in the PSP model since the metric is equivalent to FLRW.

6.3 Definition of the Phase Flow

In the PSP model:

$$\Phi_M(M) = \frac{1}{1 + z} \cdot \frac{R_{\text{shell}}(M)}{R_{\text{shell}}(0.5)} \quad (41)$$

6.4 Derivation of the Expression for $H(z)$

Using (39), (40) and (41):

$$H(z) = \frac{1}{1 + z} \cdot \frac{R_{\text{shell}}(z)}{R_{\text{shell}}(0.5)} \cdot \frac{c}{R_{\text{shell}}(z)} \cdot \frac{dR_{\text{shell}}}{dM} \cdot \frac{dM}{dz} \quad (42)$$

140 Canceling $R_{\text{shell}}(z)$:

$$H(z) = \frac{c}{R_{\text{shell}}(0.5)} \cdot \frac{1}{1 + z} \cdot \frac{dR_{\text{shell}}}{dM} \cdot \frac{dM}{dz} \quad (43)$$

But:

$$\frac{dR_{\text{shell}}}{dz} = \frac{dR_{\text{shell}}}{dM} \cdot \frac{dM}{dz} \quad (44)$$

Therefore:

$$H(z) = \frac{c}{R_{\text{shell}}(0.5)} \cdot \frac{1}{1+z} \cdot \frac{dR_{\text{shell}}}{dz} \quad (45)$$

6.5 Connection to the Calibration Expression

From DESI and Pantheon+ data, it is derived:

$$\frac{R_{\text{shell}}(z)}{R_{\text{shell}}(0.5)} = \sqrt{1 + 1.25 \left(\frac{z}{2.5}\right)^2 e^{0.35z}} \quad (46)$$

145 Substituting (46) into (45) and computing the derivative, we obtain:

$$H(z) = H_0 \sqrt{1 + \alpha \left(\frac{z}{2.5}\right)^2 e^{\beta z}} \quad (47)$$

where:

$$H_0 = \frac{c}{R_{\text{shell}}(0.5)}, \quad \alpha = 0.15, \quad \beta = 0.35 \quad (48)$$

Conclusion: The expression for $H(z)$ naturally follows from the PSP geometry.

7 Dimensional Analysis

7.1 Check of Equation (31)

$$\Phi_M^2 \left(\frac{1}{a} \frac{da}{dM} \right)^2 = \frac{8\pi G}{3c^2} \rho(M) - \frac{k}{a^2}$$

150 **Dimensional analysis:**

- Φ_M — dimensionless,
- $(1/a \cdot da/dM)$ — $1/L$,
- $\Phi_M^2 (1/a \cdot da/dM)^2$ — $1/L^2$.

Right-hand side:

- 155
- $G\rho/c^2$ — $(L^3/(MT^2)) \cdot (M/L^3)/(L^2/T^2) = 1/L^2$,
 - k/a^2 — $1/L^2$.

Dimensions match. The equation is correct.

7.2 Check of Equation (39)

$$H(M) = \Phi_M(M) \cdot \frac{c}{R_{\text{shell}}(M)} \cdot \frac{dR_{\text{shell}}}{dM}$$

Dimensional analysis:

- 160
- H — $1/T$,

- Φ_M — dimensionless,
- c/R — $(L/T)/L = 1/T$,
- dR/dM — dimensionless.

Dimensions match. The equation is correct.

165 8 Physical Interpretation

8.1 Equivalence of PSP and FLRW

It has been established that the PSP metric is **equivalent** to the FLRW metric after a change of the time coordinate. This means that:

1. The PSP model does not contradict general relativity.
- 170 2. All standard cosmological tests (supernovae, BAO, CMB) are applicable to PSP.
3. The differences between PSP and Λ CDM arise from the **dynamics of the phase flow** $\Phi_M(M)$ and the **geometry of $R_{\text{shell}}(M)$** .

8.2 Phase Flow as a Source of Acceleration

In standard Λ CDM, the accelerated expansion is driven by the cosmological constant Λ . In the
175 PSP model, acceleration arises from:

1. **Centrifugal acceleration** (torus rotation) — described by the parameter α .
2. **Gravitational gradient** (SMBH displacement) — described by the parameter β .

This means that dark energy in PSP is **not an entity**, but a **geometric effect**.

8.3 Local Limit

180 In the local limit $M \rightarrow 0$:

- $\Phi_M \rightarrow 1$,
- $R_{\text{shell}} \rightarrow R_0$,
- $dR_{\text{shell}}/dM \rightarrow 0$.

Equation (31) reduces to:

$$0 = \frac{8\pi G}{3c^2} \rho - \frac{k}{R_0^2} \quad (49)$$

185 This corresponds to a **static solution** — no expansion in the local region. The metric reduces to the Minkowski metric, consistent with the equivalence principle.

9 Conclusion

The following results have been obtained in this document:

1. **The PSP metric is equivalent to the FLRW metric** after the change of time coordinate $t \leftrightarrow M$.

2. **The Friedmann equations were derived in the phase representation:**

$$\Phi_M^2 \left(\frac{1}{R_{\text{shell}}} \frac{dR_{\text{shell}}}{dM} \right)^2 = \frac{8\pi G}{3c^2} \rho(M) - \frac{k}{R_{\text{shell}}^2} \quad (50)$$

$$\Phi_M^2 \frac{1}{R_{\text{shell}}} \frac{d^2 R_{\text{shell}}}{dM^2} + \Phi_M \frac{d\Phi_M}{dM} \frac{1}{R_{\text{shell}}} \frac{dR_{\text{shell}}}{dM} = -\frac{4\pi G}{3c^4} (\rho c^2 + 3p) \quad (51)$$

3. **The expression for the Hubble parameter was derived:**

$$H(z) = \Phi_M(z) \cdot \frac{c}{R_{\text{shell}}(z)} \cdot \frac{dR_{\text{shell}}}{dM} \cdot \frac{dM}{dz} \quad (52)$$

4. **It was shown that the empirical expression:**

$$H(z) = H_0 \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2} e^{\beta z} \quad (53)$$

is a consequence of the PSP geometry, not a fitting parameter.

5. **Dimensional analysis was performed**, confirming the correctness of all equations.

Main conclusion: The PSP model is a **natural generalization** of the standard cosmological model, following from general relativity and the geometry of a closed cycle.

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Appendix A: Detailed Derivation of Einstein's Field Equations

A.1 FLRW Metric

$$ds^2 = -c^2 dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right)$$

A.2 Christoffel Symbols

$$\Gamma_{ij}^0 = \frac{\dot{a}}{c} a \gamma_{ij}, \quad \Gamma_{0j}^i = \frac{\dot{a}}{c} \delta_j^i$$

A.3 Ricci Tensor

$$R_{00} = -\frac{3}{c^2} \ddot{a}, \quad R_{ij} = \frac{1}{c^2} (2\dot{a}^2 + a\ddot{a} + 2kc^2) \gamma_{ij}$$

A.4 Scalar Curvature

$$R = \frac{6}{c^2} \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2} \right)$$

220 A.5 Einstein Tensor

$$G_{00} = \frac{3}{a^2} (\dot{a}^2 + kc^2), \quad G_{ij} = -\frac{1}{a^2} (2a\ddot{a} + \dot{a}^2 + kc^2) \gamma_{ij}$$